REPORT

ON

"ANALYSIS OF SOIL SLOPE STABILITY WITH AND WITHOUT REINFORCEMENT USING MIDAS GTS NX"

SUBMITTED BY

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INTRODUCTION

Soil slopes are an important part of many civil engineering projects, such as highways, railways, embankments, dams, and excavations. The stability of a slope depends on several factors, including soil properties, slope geometry, groundwater conditions, applied loads, and external disturbances. If a slope becomes unstable, it can lead to landslides, excessive deformation, damage to nearby infrastructure, and significant economic losses. Therefore, studying the stability and deformation of soil slopes is an important aspect of geotechnical engineering.

One of the common ways to improve slope stability is to modify the slope geometry or provide reinforcement to increase the shear resistance of the soil. Reinforcement systems such as geogrids, geotextiles, soil nails, and other reinforcement elements are widely used for this purpose. However, their effectiveness depends on factors such as reinforcement strength and stiffness, spacing, arrangement, and the interaction between the soil and reinforcement.

Traditional slope stability methods are generally based on simplified assumptions and limit equilibrium concepts. While these methods are useful for basic stability assessment, numerical methods such as the Finite Element Method (FEM) can provide a more detailed understanding of the stress, strain, displacement, and overall deformation behaviour of a slope. **MIDAS GTS NX**, a finite-element-based geotechnical analysis software, can be used to model complex soil behaviour and soil–reinforcement interactions under different conditions.

In this study, a soil slope is analysed using **MIDAS GTS NX** for two different conditions: **without reinforcement and with reinforcement**. First, the unreinforced slope is modelled to study its stability and deformation behaviour. Reinforcement is then added to the slope, and the results are compared with those of the unreinforced condition. Parameters such as **factor of safety, displacement, stress distribution, and deformation patterns** are considered to understand the effect of reinforcement on slope performance.

The study helps in understanding how reinforcement can improve the stability and reduce the deformation of soil slopes. It also demonstrates how finite element analysis can be applied to investigate practical geotechnical engineering problems and assist in selecting suitable reinforcement strategies for safer and more stable slopes.

OBJECTIVES

The major objectives are:

1. To analyze the Soil Slope with and without reinforced.
2. To analyze load, pour water pressure and pseudo analysis.
3. To analyze factor of safety, displacement, strain, stress strain and grid force with graph.

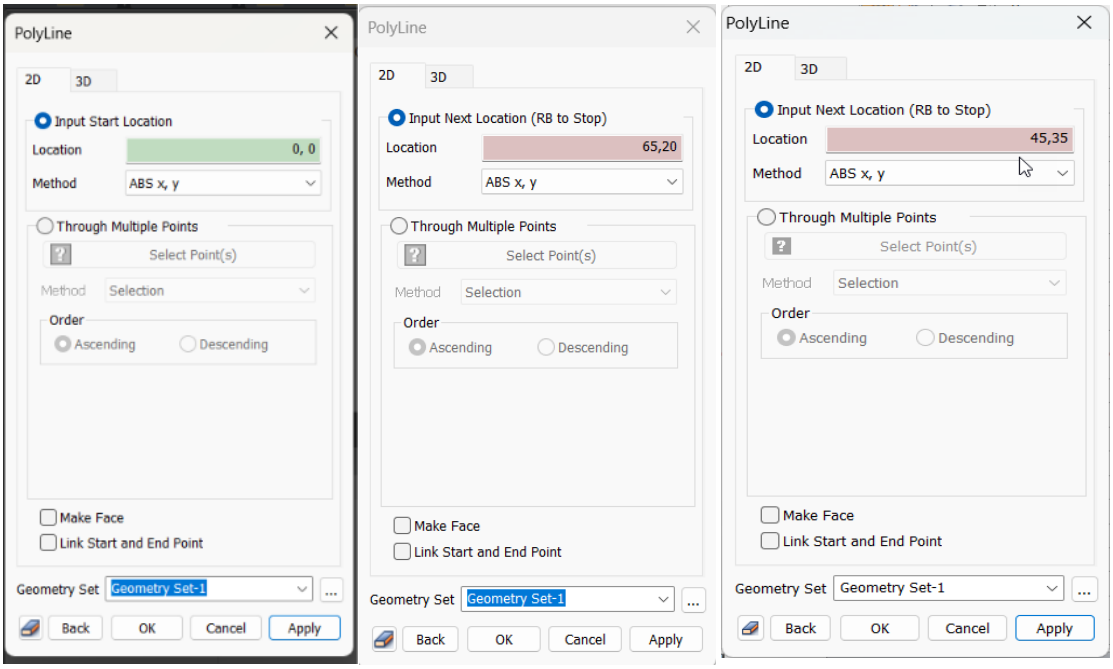
GEOMETRY

➢ CREATING A POLYLINE-FOR SLOPE

- STEP 1:
  - ➢ Go to Geometry> Polyline

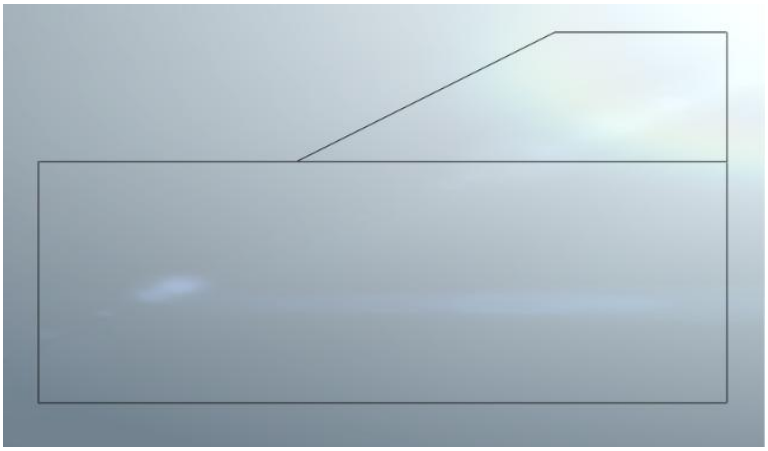


- STEP 2:
  - ➢ Select method 2D, Input One corner at Location 0,0 (Abs X,Y).
  - ➢ Input Next location 65,0 (Abs X,Y), Then click apply.
  - ➢ Input Next location 65,20 (Abs X,Y), Then click apply.

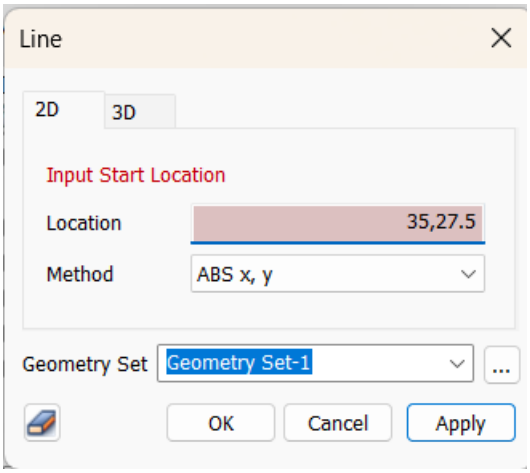


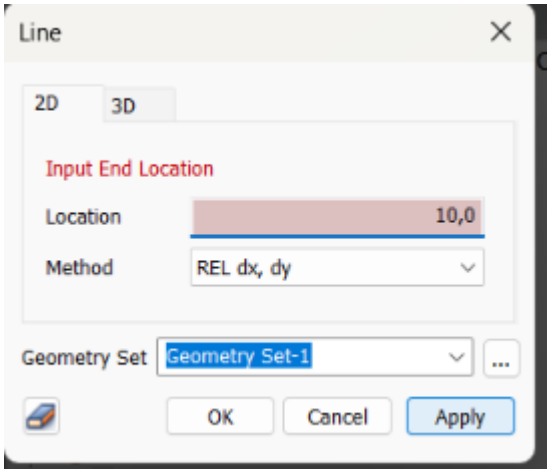
  - ➢ Input Next location 0,20 (Abs X,Y), Then click apply.
  - ➢ Input Next location 0,0 (Abs X,Y), Then click apply.

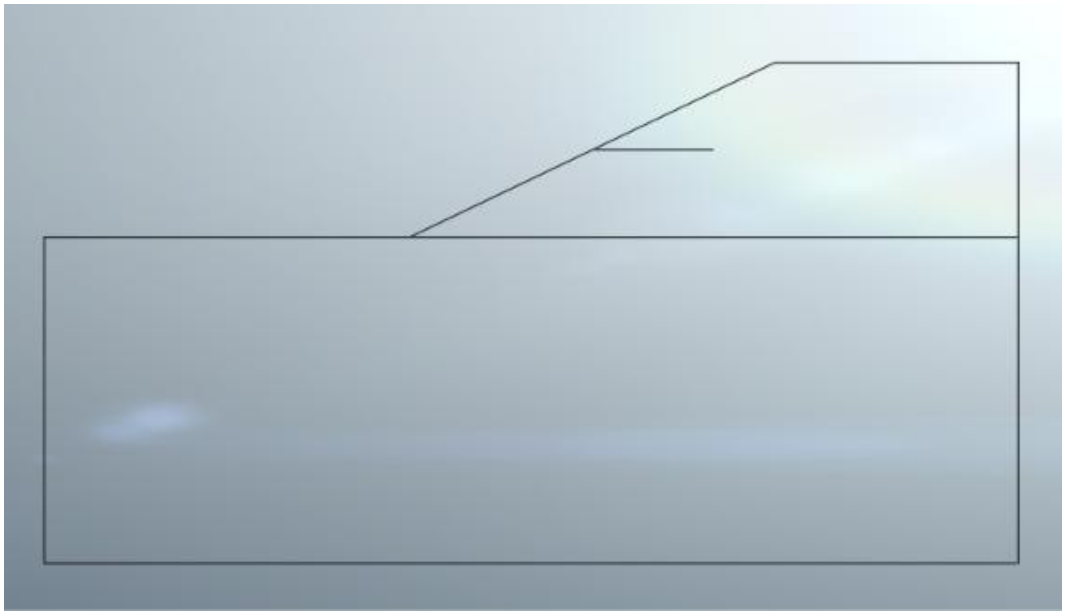
- Input Next location 65,20(Abs X,Y), Then click apply.
- Input Next location 65,35 (Abs X,Y), Then click apply.
- Input Next location 45,35 (Abs X,Y), Then click apply.
- Input Next location 25,20 (Abs X, Y), Then click apply.



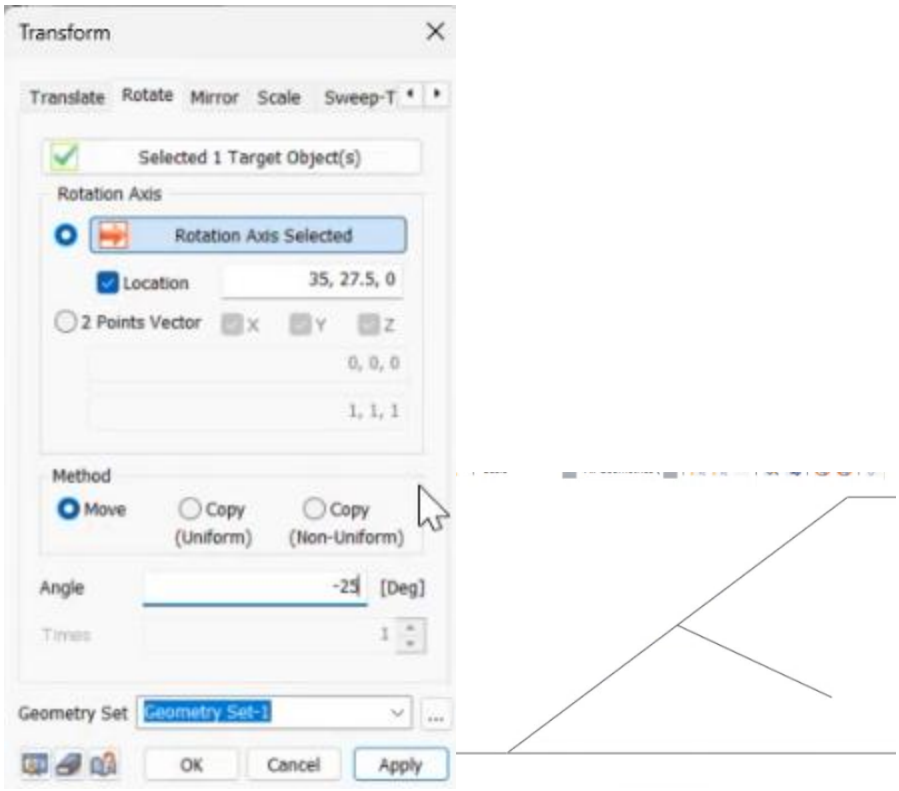
- STEP 3: Defining Nails
  - Go to Geometry>line.
  - Create the line with shown coordinates with nail length of 10m.



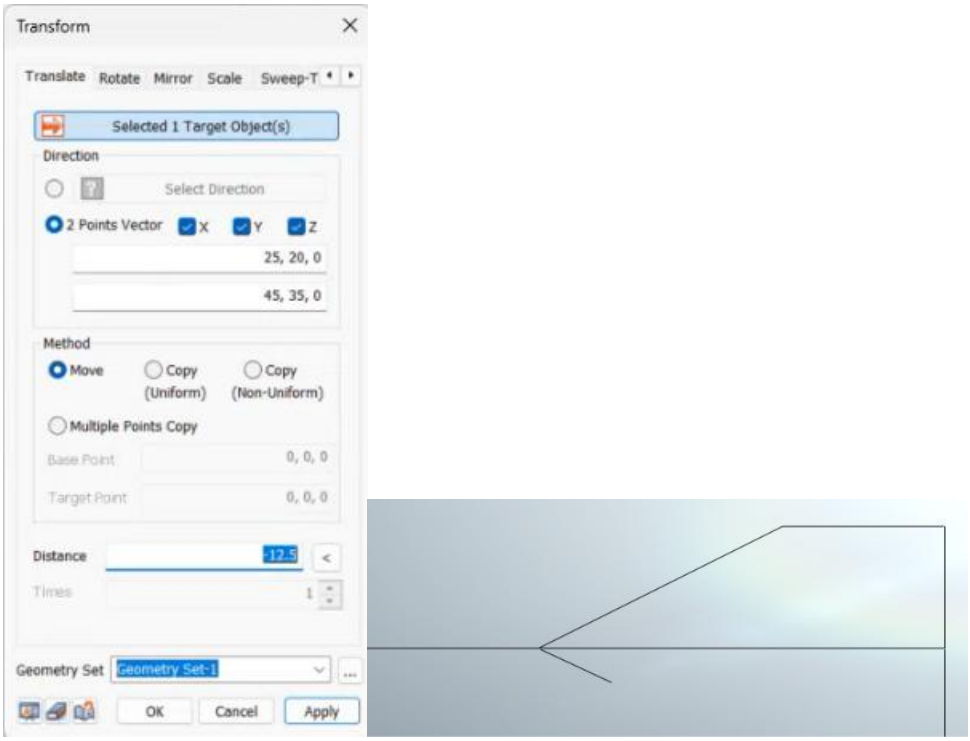




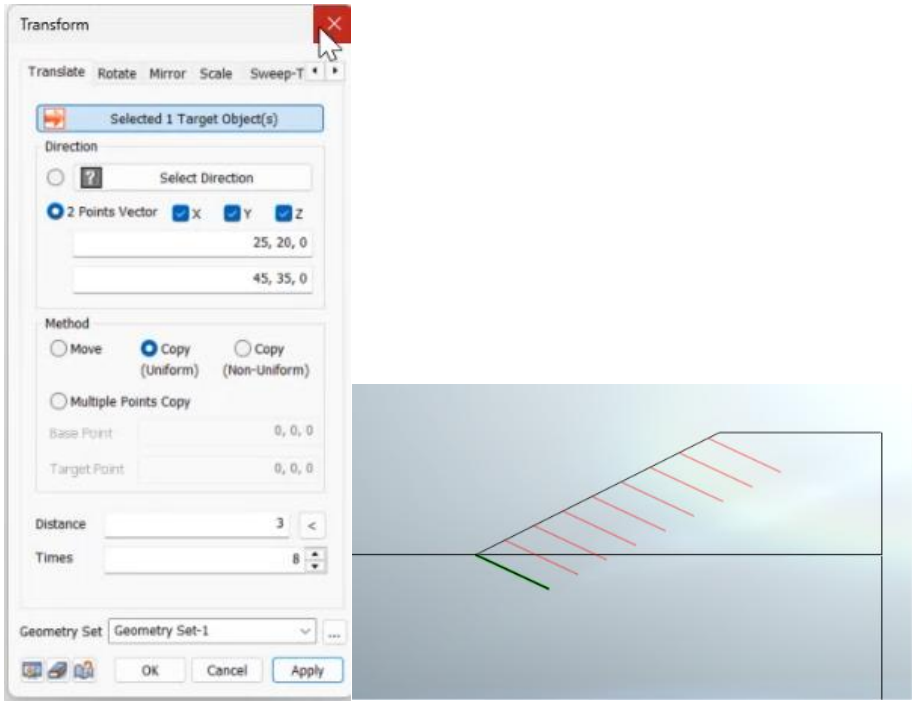
- Go to Geometry>Transform>Rotate.
- Select the target object as nail.
- Select Rotation axis as 2-point vector> Define first point as starting point of nail > next point rotate with respect to Z-axis as depicted below.
- Angle is given as -25 degree
- See preview before proceeding> Click OK.



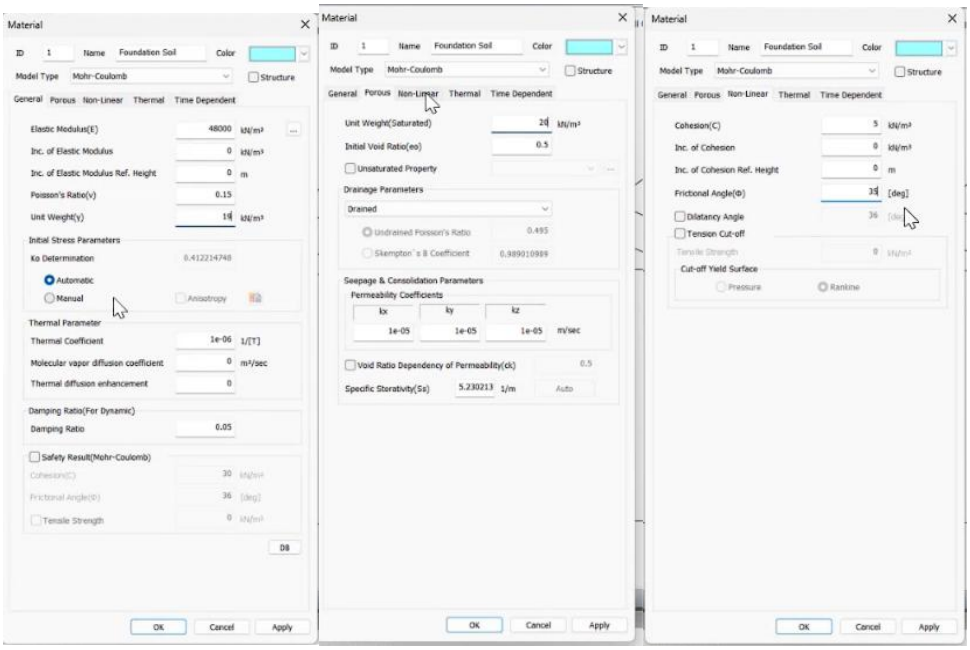
- Go to Geometry>Transform>Translate.
- Select target object as nail.
- Define 2 points vector as shown in in picture.
- Define Move distance as 16.8m.
- See preview before proceeding to next step.
- Then Apply.



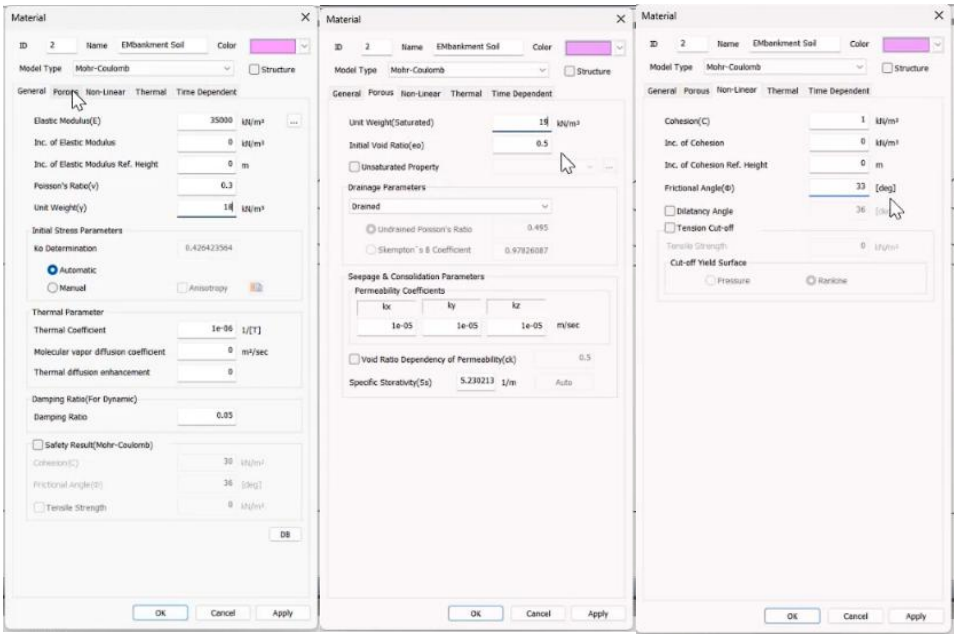
- Go to Geometry>Transform>Translate.
- Select target object as nail
- Define 2 points vector as shown in in picture.
- Copy at a distance of 3m, 8 times.
- See preview before proceeding to next step
- Then Apply. Delete unnecessary lines.



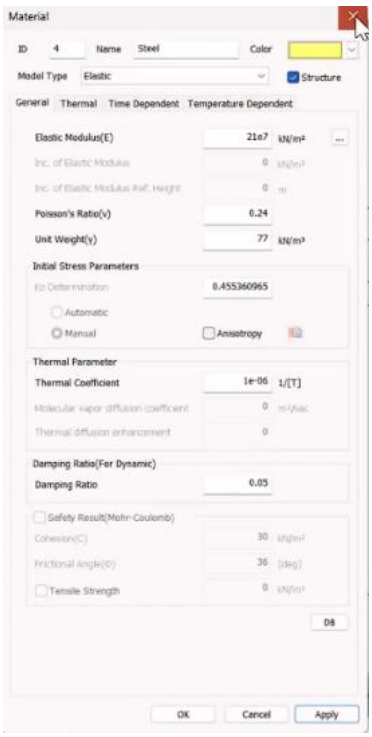
- STEP 4: Defining Material
    - Go to Mesh > Material
    - Click on Create. Select Isotropic
    - Select Model type as Mohr-Coulomb
    - In General tab, Enter the value of Elastic modulus (E) =48000 kN/m2
    - Enter Poisson's ratio= 0.15, Unit weight = 19 kN/m3.
    - Give name as Foundation Soil.
    - In Porous tab, Enter the value of Unit Weight (Saturated)=20kN/m2
    - In non-linear tab, Enter the value of Cohesion(C)=5kN/m2, Frictional Angle=35[deg].



- In General tab, Enter the value of Elastic modulus (E) =35000 kN/m2
- Enter Poisson's ratio= 0.3, Unit weight = 18 kN/m3.
- Give name as Embankment Soil.
- In Porous tab, Enter the value of Unit Weight (Saturated)=19KN/m2
- In non-linear tab, Enter the value of Cohesion(C)=1kN/m2, Frictional Angle=33[deg].

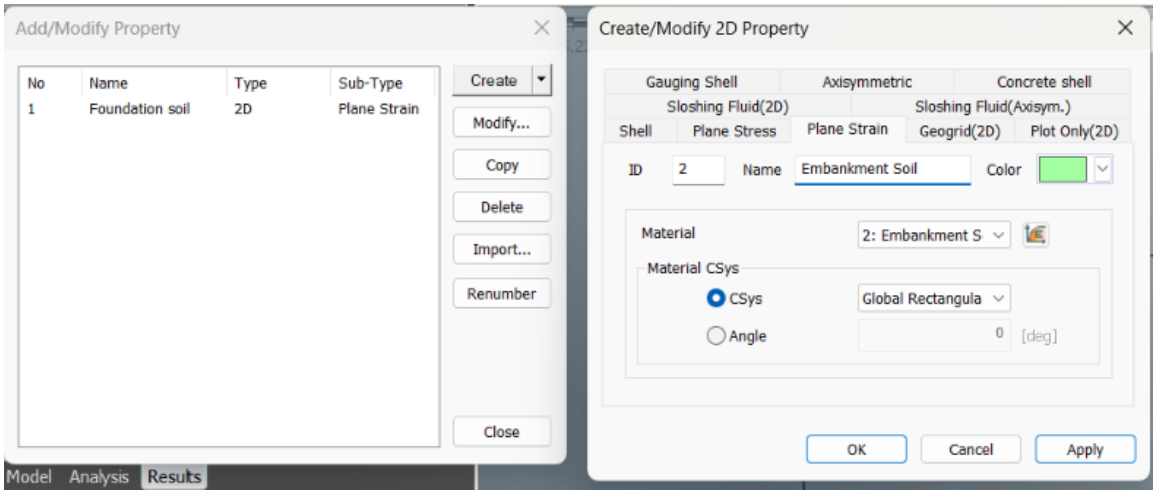


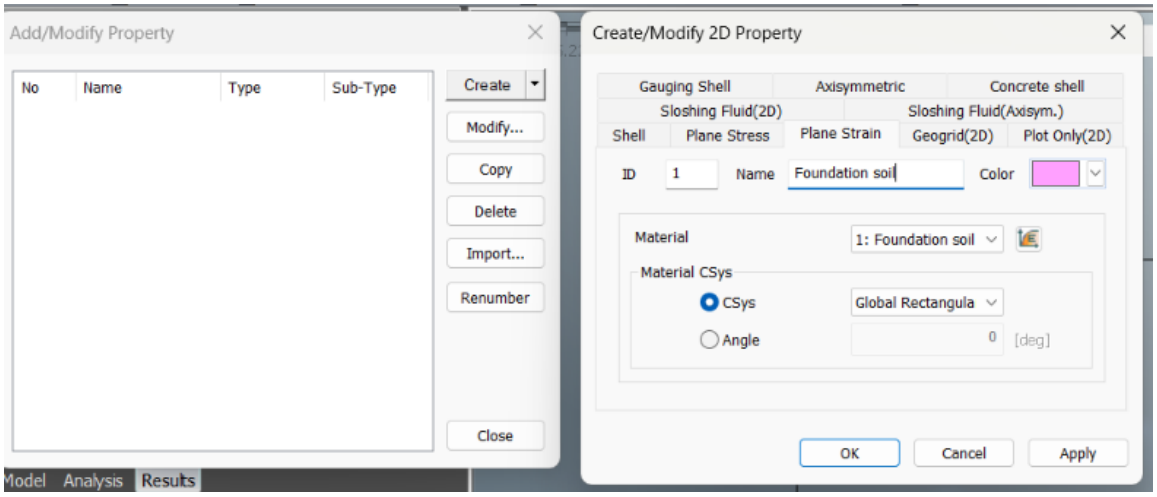
- In General tab, Enter the value of Elastic modulus (E) =21e7 kN/m2
- Give name as Steel.



- STEP 5: Defining Property
  - Go to Mesh>Click on Property

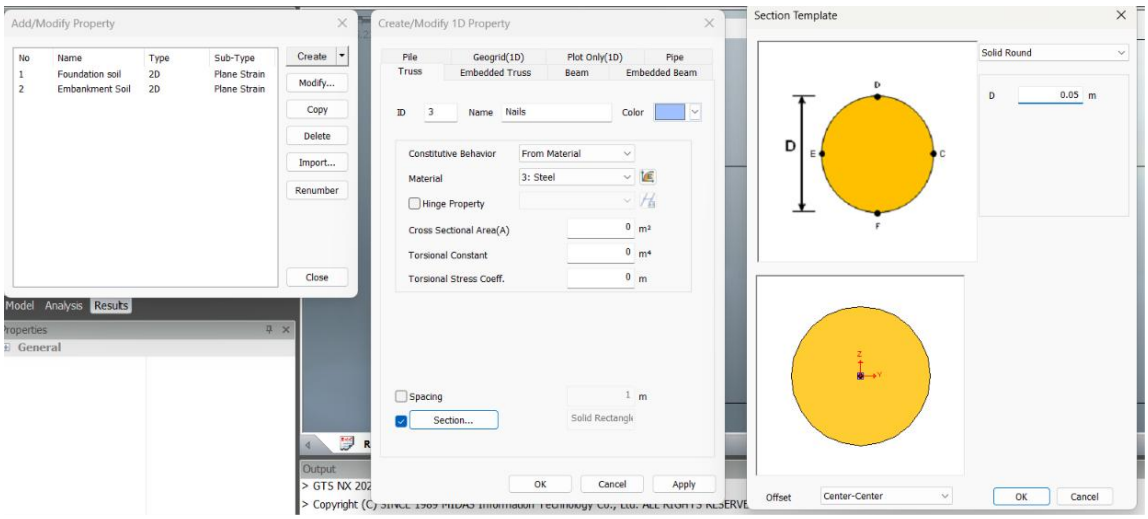
- Click on Create. Select 2D
- Select Plane Strain
- Name it as Embankment SOIL
- Select Embankment SOIL from Material drop- down menu.
- Click Apply.
- Similarly Define the 2D property for Foundation layer.





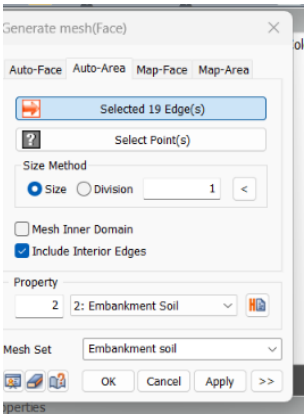
- Go to Mesh>Click on Property
- Click on Create. Select 1D

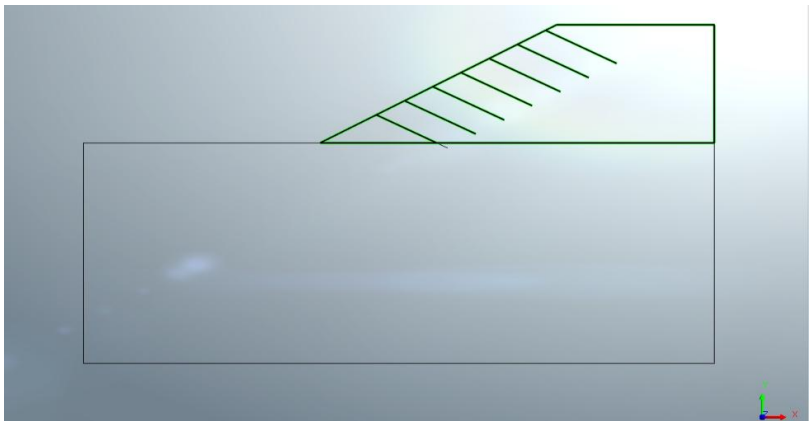
- Click on Truss.
- Name it as Nail.
- Select Material as Nail. Check on section. Select the type of the section as Solid Round and
- Enter the diameter as 0.05 m
- Click OK



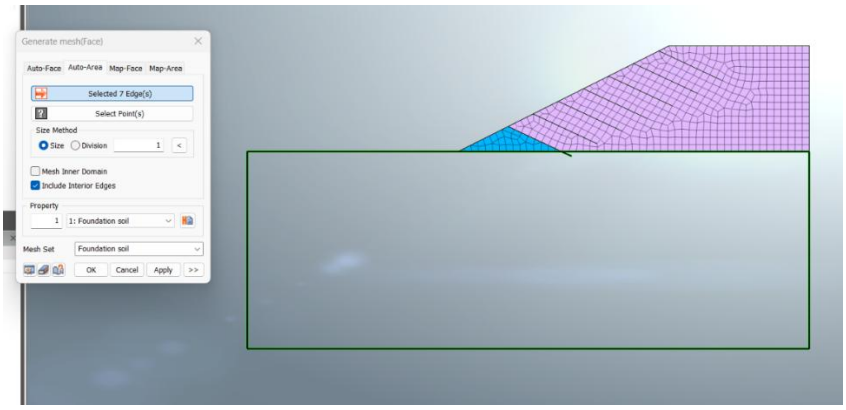
MESHING

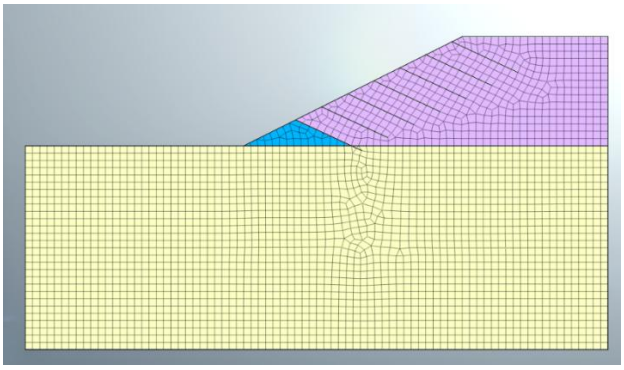
- STEP 1:
  - Go to Mesh>Generate>2D
  - Select the lines as shown.
  - Enter the size as 1 m. Tick on the “Mesh Inner Domain”.
  - The property is Embankment Soil.
  - Give the Mesh set name “Embankment Soil”.
  - Enable the Higher Order Element and Midside Nodes on Geometry, Click Apply.





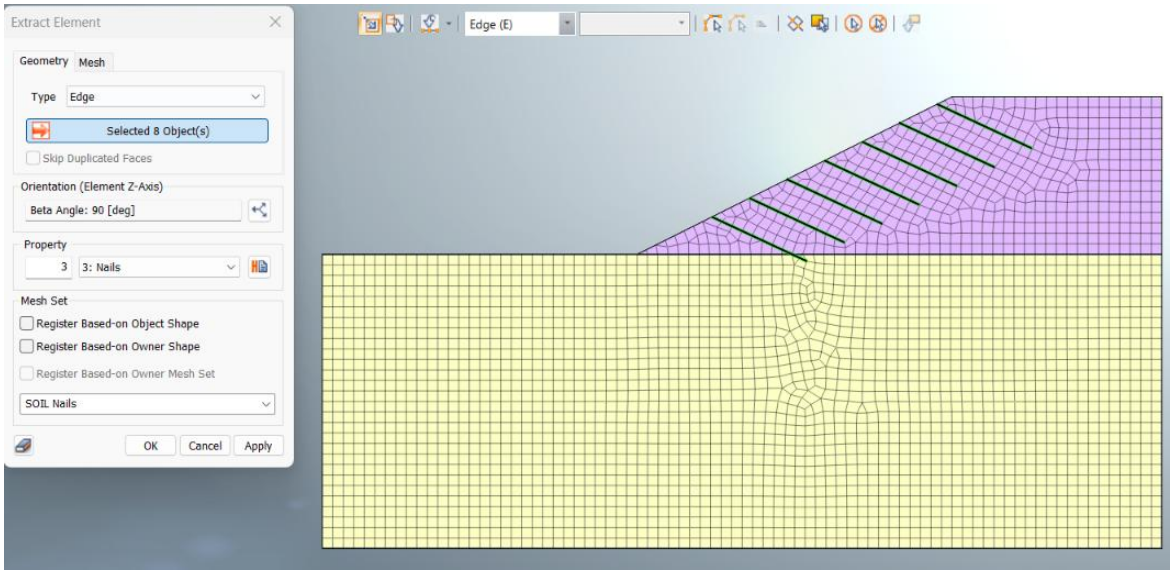
- Go to Mesh>Generate>2D
- Select the lines as shown.
- Enter the size as 1 m. Tick on the “Mesh Inner Domain”. The property is Foundation Soil.
- Give the Mesh set name “Foundation Soil”.
- Enable the Higher Orderm Element and Midside Nodes on Geometry. Click Apply.





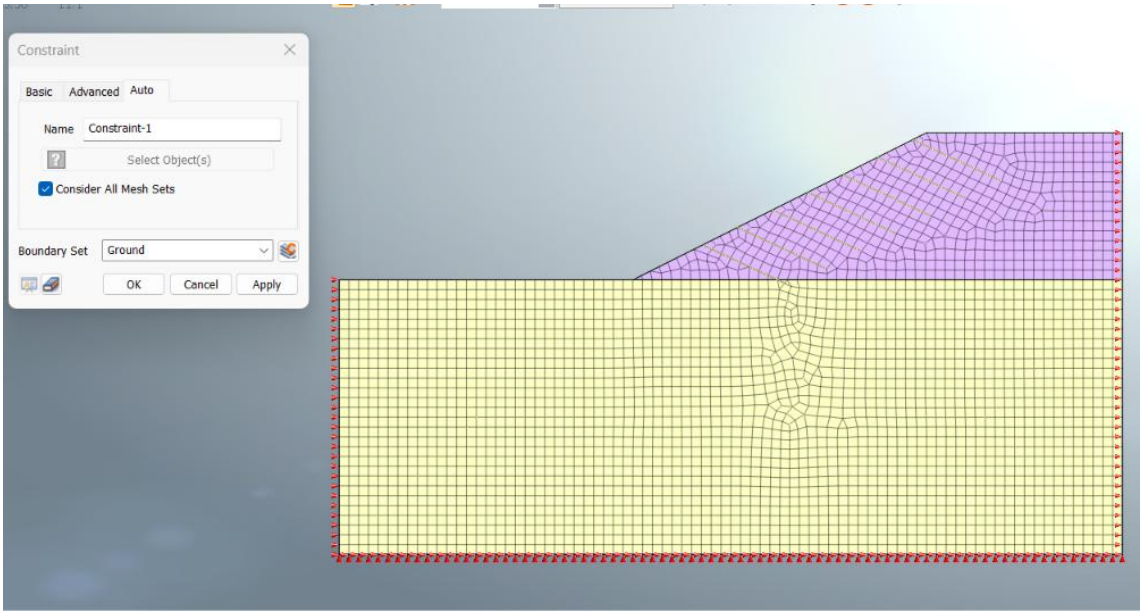
- Merge mesh 2 and 3.

- Go to Mesh>Element>Extract
- Change the type to Edge.
- Select the all nails as shown
- The property is Nail.
- Name mesh set as Nail.
- Click OK.

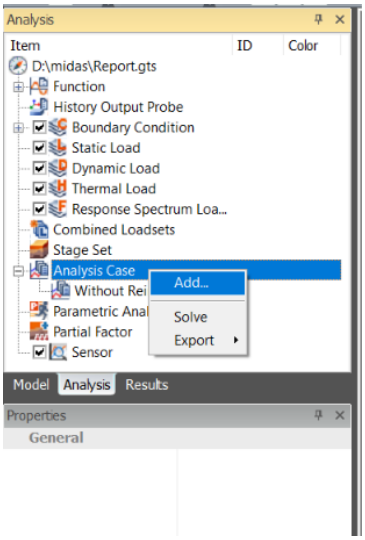


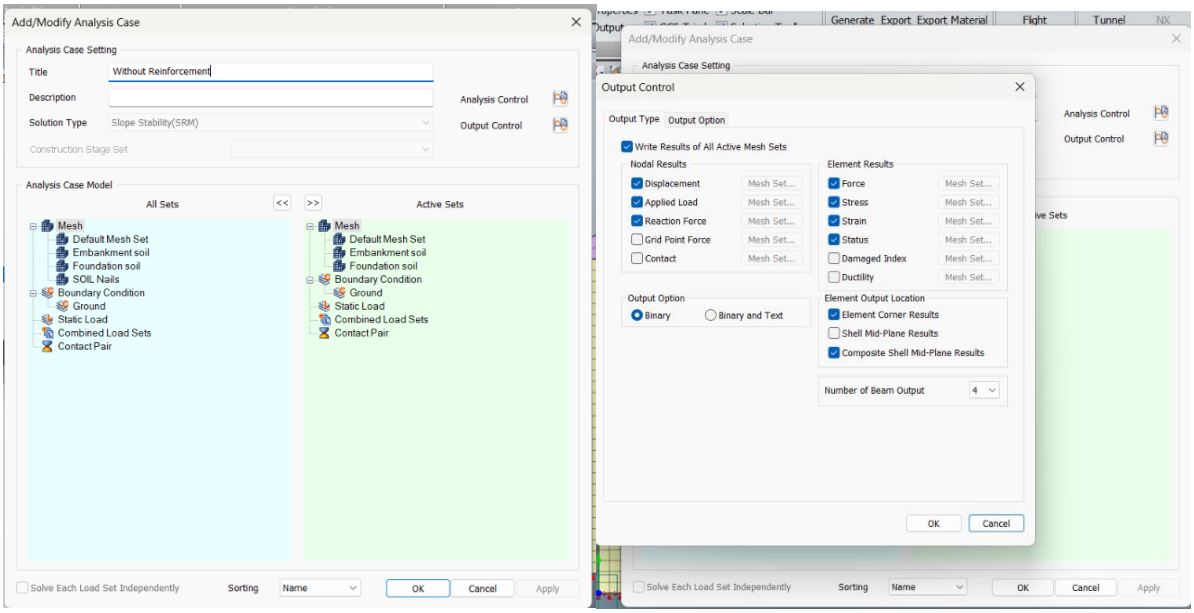
BOUNDARY CONDITION

- STEP 1:
  - Go to Static/Slope Analysis> Boundary > Constraint.
  - Select Auto, check on “Consider All mesh Sets”.
  - Name the boundary set as “Ground”.
  - Click OK.

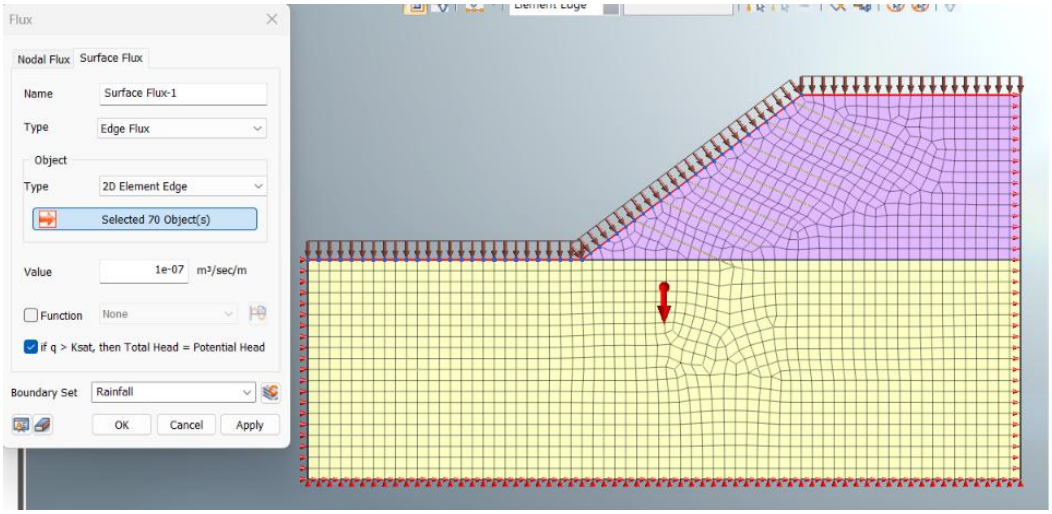


- STEP 2:
  - Go to Analysis > Analysis case
  - Right click Analysis case and then select Add.
  - Name the title as “Without Reinforcement” and solution type as “Slope stability (SRM)”.
  - Remove “Soil Nails” from active sets and add “Ground” from all sets to active sets.
  - Click “output control” and select “strain” from element result.
  - Click OK.

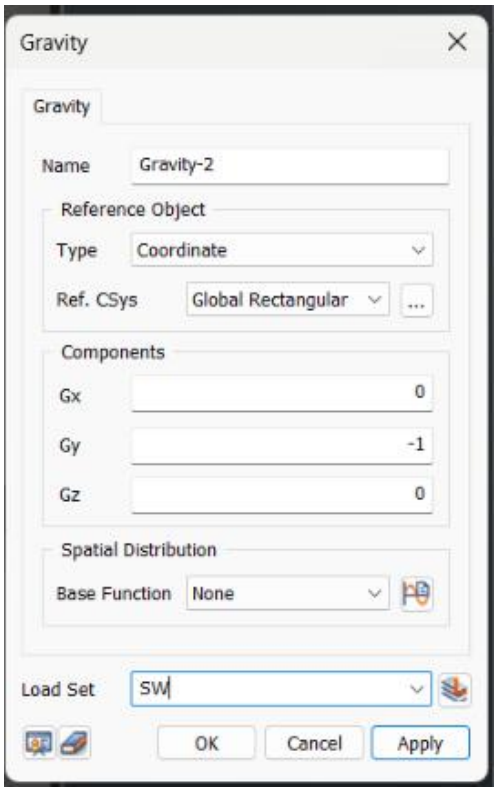




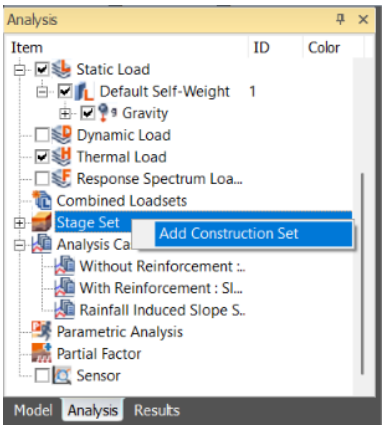
- STEP 4:
  - Repeat the step again and just add the “Soil Nails” from active sets and add “Ground” from all sets to active sets.
  - Name the title as “With Reinforcement” and solution type as “Slope stability (SRM)”.
  - Click “output control” and select “strain” from element result.
  - Click OK.
- STEP 5: Defining Rainfall
  - Go to Seepage/Consolidation Analysis > Boundary > Surface Flux
  - Select Type as Edge Flux. Select 2D Element Edge as highlighted in Red Color.
  - Enter Value as 1e-07 m3/sec/m
  - Check in if q > Ksat, then Total Head = Potential Head.
  - Name Boundary set as Rain Fall.
  - Click OK.

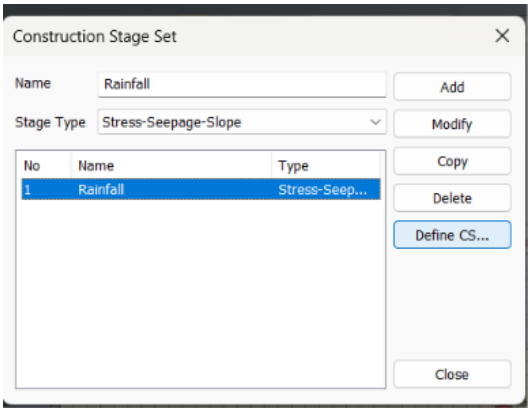


- STEP 6: Self Weight
  - Go to Seepage/Consolidation Analysis > Load >Self Weight.
  - Check for gravity Direction, Gy should be -1.
  - Name the Load set as “SW”.
  - Click “OK”.

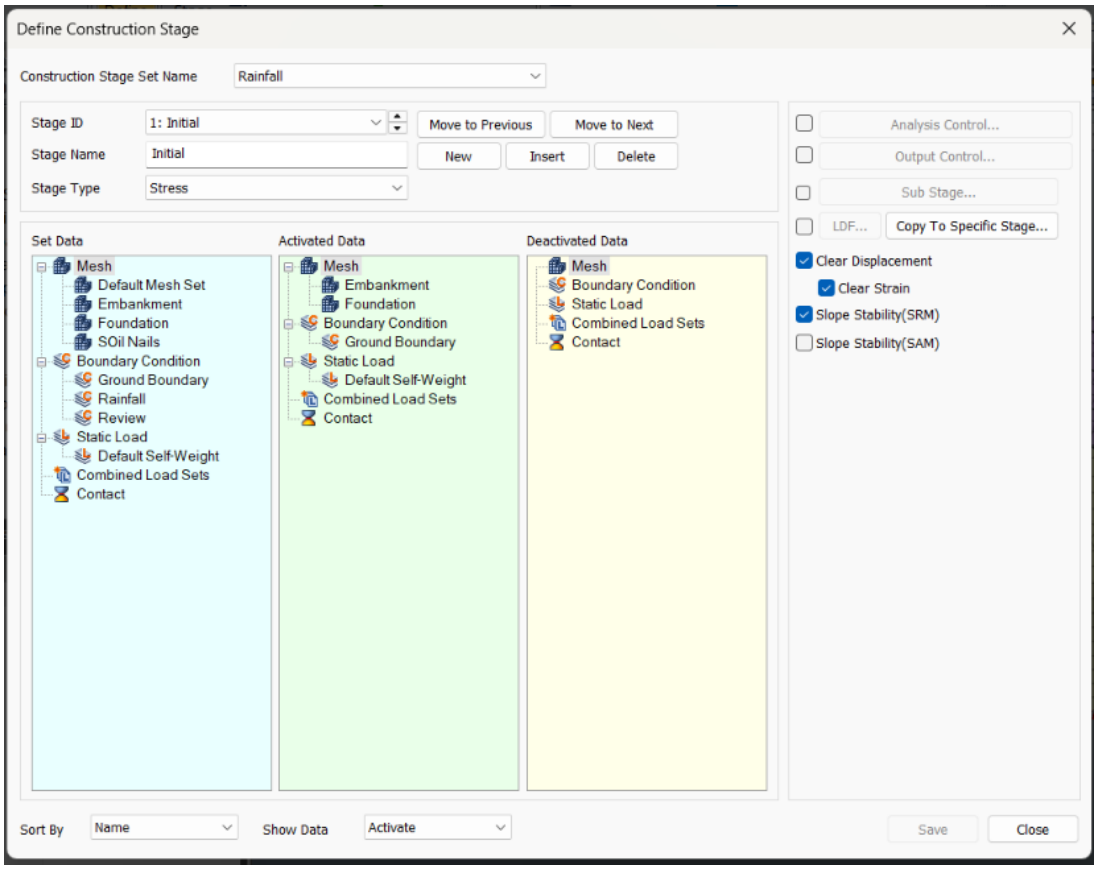


- STEP 7:
  - In works-tree go to Analysis tab.
  - Right click on Stage Set. Then click on Add Construction Stage.
  - Select Stage Type as Stress-Seepage-Slope.
  - Enter name as “Rain Fall”
  - Click Add
  - Double Click on Rain Fall.

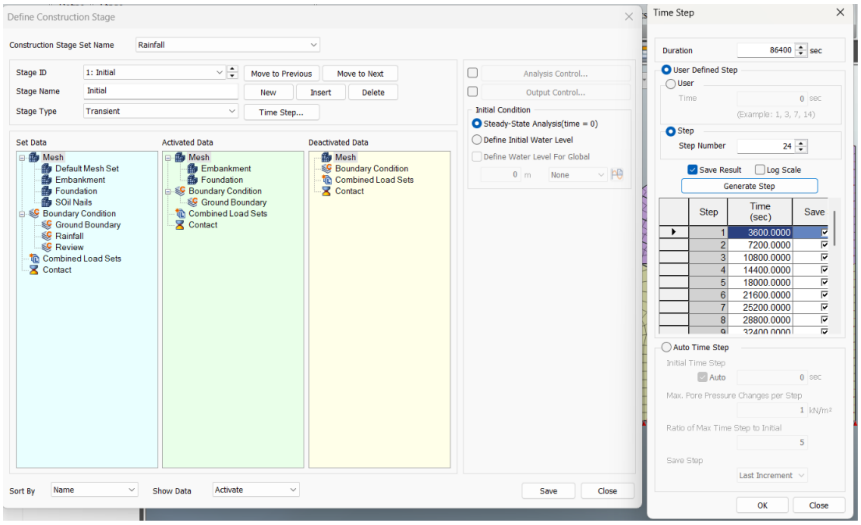




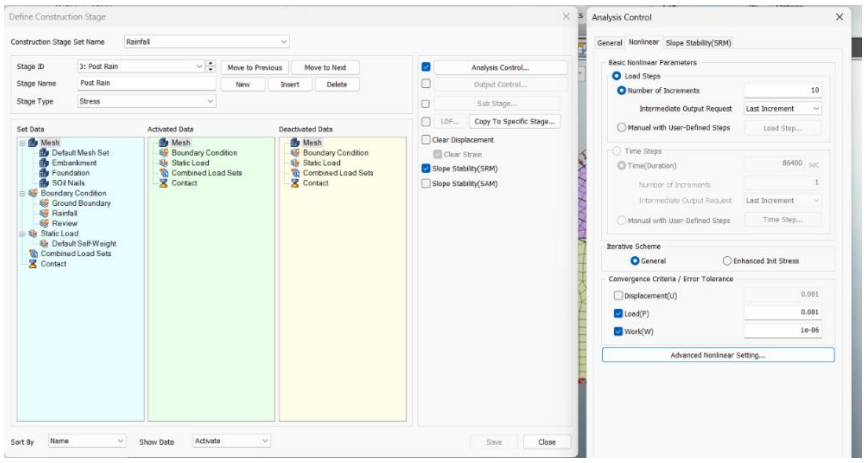
- STEP 8:
  - Enter Stage Name as Initial.
  - Select Stage Type as Stress. Activate “Embankment Soil, Foundation Soil” and “Ground” Boundary Condition and “SW” Static Load by Drag and Drop method.
  - Check Clear Displacement.
  - Check Slope Stability (SRM).
  - Click Save.
  - Click New.



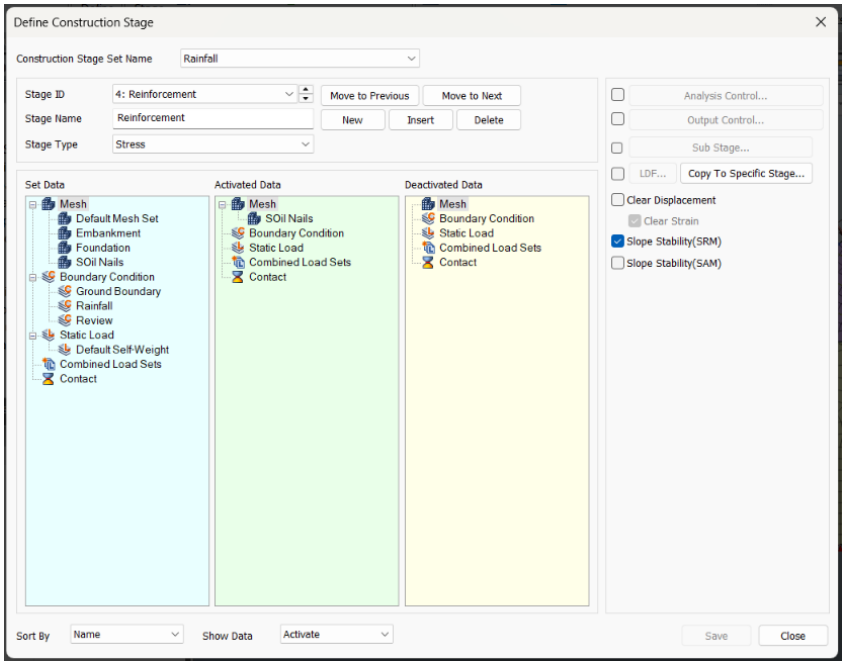
- STEP 9:
  - Enter Stage Name as Rain Fall.
  - Select Stage Type as Transient. Activate “Rain Fall and Review” by Drag and Drop method.
  - Click Time Step.
  - Enter Time Step Duration as 86400 sec.
  - Enter Step Number as 24.
  - Check Save Result and click on Generate Step.
  - Click OK
  - Click Save
  - Click New



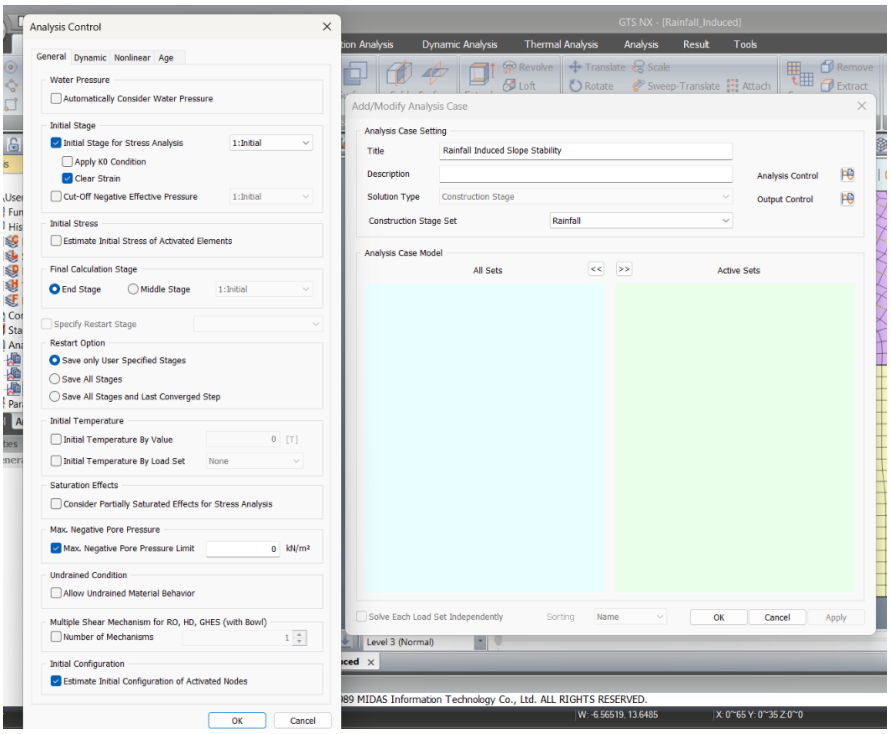
- STEP 10:
  - Enter Stage Name as Post Rain.
  - Select Stage Type as Stress.
  - Click on Analysis Control.
  - Select Nonlinear tab.
  - In Load Steps enter Number of Increments as 10 and in Intermediate Output Request select Every Increment.
  - Click OK
  - Check Slope Stability (SRM)
  - Click Save.
  - Click New.



- STEP 11:
  - Enter Stage Name as Nail. Select Stage Type as Stress. Activate "Nail" by Drag and Drop method.
  - Check Slope Stability (SRM)
  - Click Save.
  - Click Close.



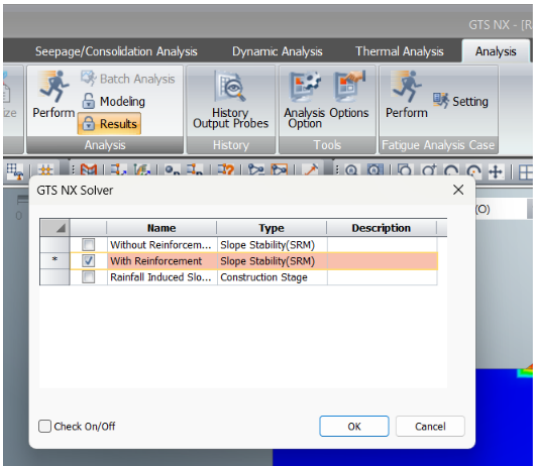
- STEP 12:
  - Go to Analysis > Analysis case
  - Right click Analysis case and then select Add.
  - Name the title as "Rainfall" and solution type as "Construction Stage".
  - Click "Analysis Control" and select "strain, Estimate initial configuration of activated nodes" from Analysis Control.
  - Click "output control" and select "strain" from element result.
  - Click OK.
  - Click Apply.



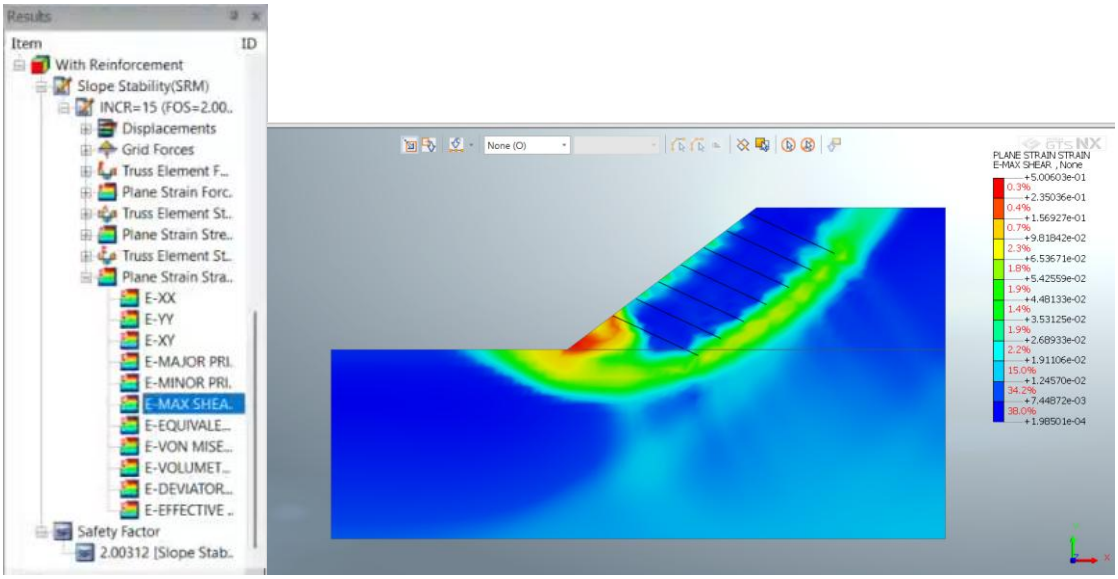
ANALYSIS:

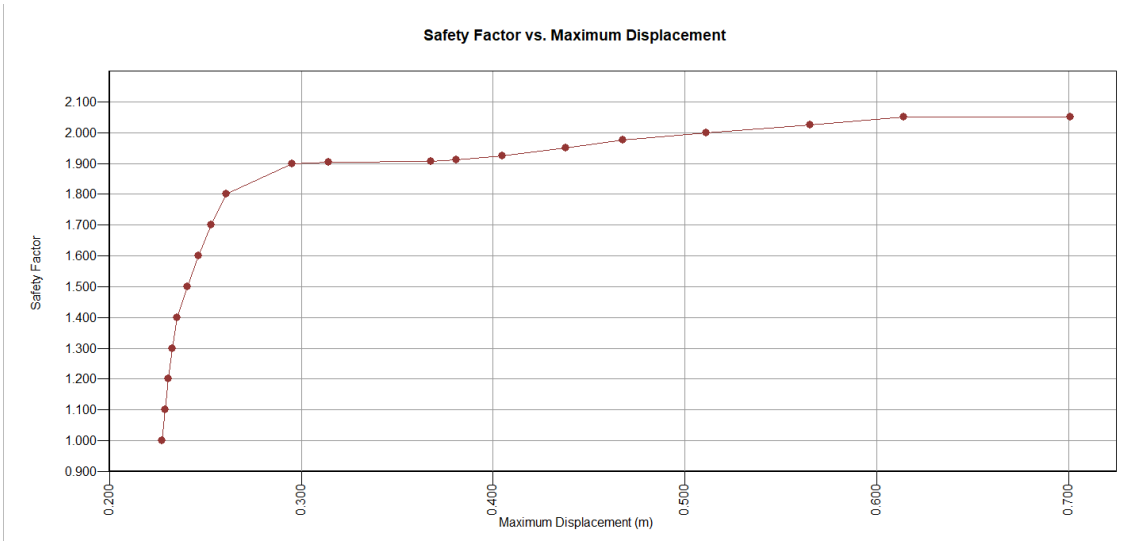
CASE 1: WITH REINFORCEMENT

- STEP 1:
  - Go to Analysis>Perform.
  - Select “With Reinforcement” from GTS NX solver.
  - Click Run.

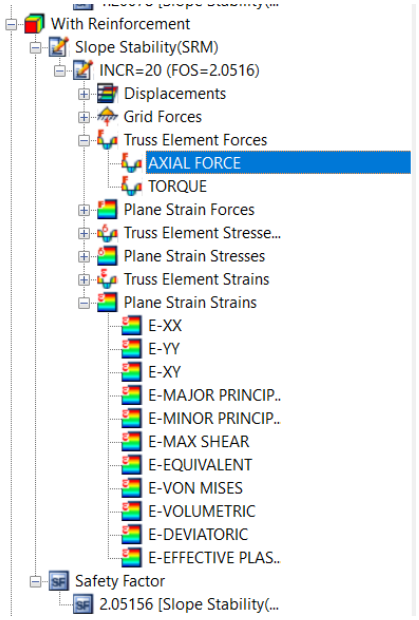


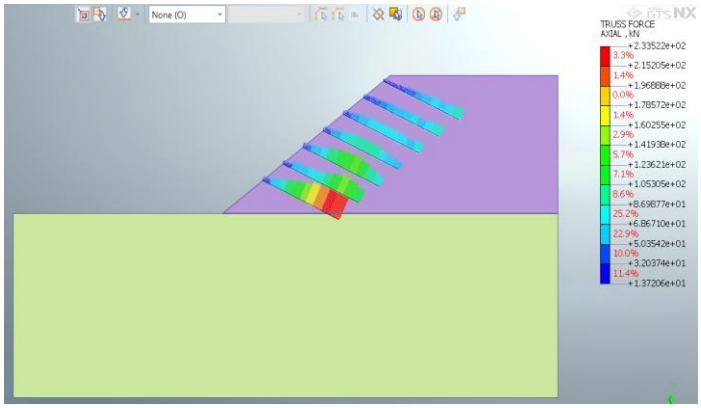
- STEP 2:
  - Go to Results > With Reinforcement
  - Click on plane strain Strains
  - One can see the localization of strains by double clicking on the E-Max shear tab.
  - The Safety Factor tab shows the calculated critical factor of safety of slope i.e. safety factor = 2.05156.





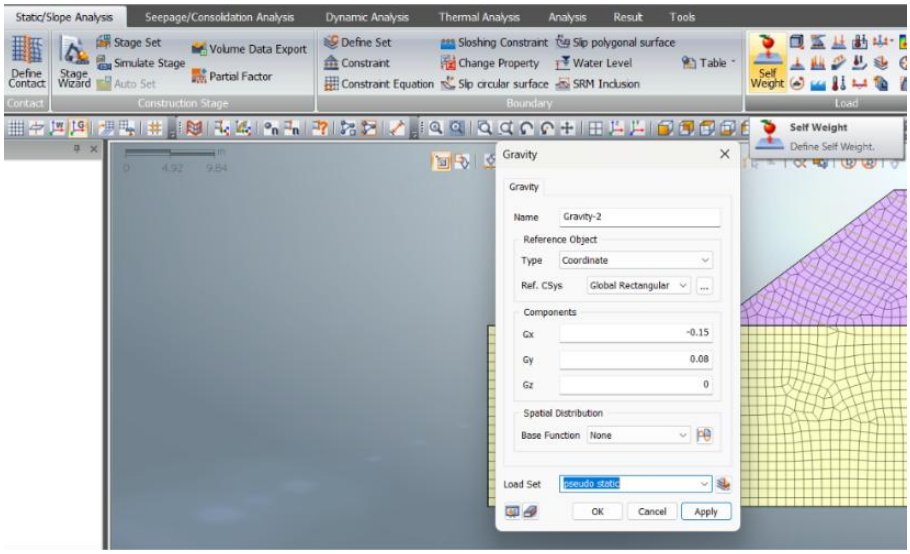
- STEP 3:
  - Go to Results >Reinforcement-SRM
  - Click on Truss Element Forces.
  - Click on the Axial Forces to check forces on the individual nails.



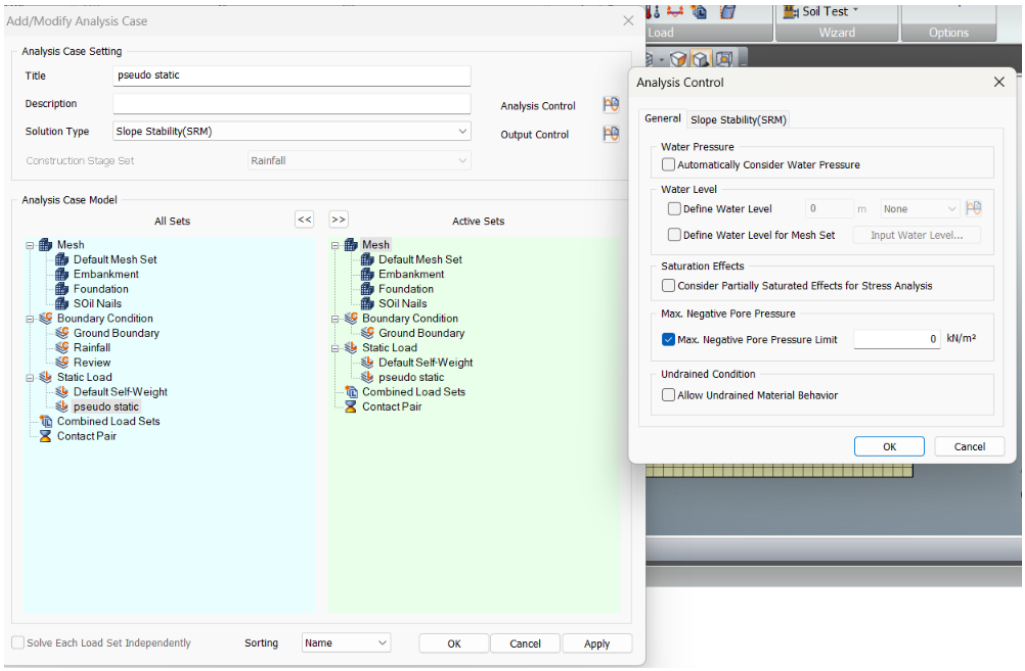


PSEUDO ANALYSIS

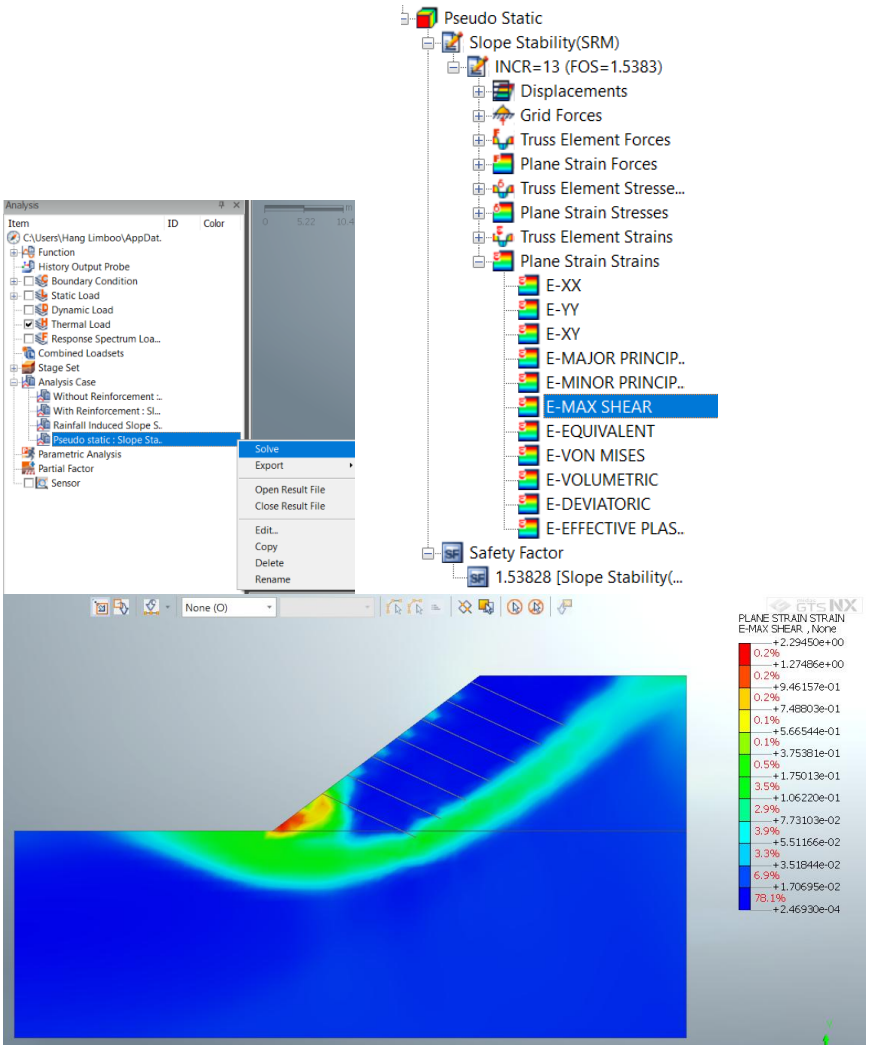
- STEP 1:
  - Go Static/slope analysis>self-weight.
  - Under Gravity>components, enter the values of Gx = -0.15 and Gy=0.07.
  - And name the load set as “pseudo static”
  - Click apply.



- STEP 2:
  - Go to Analysis > Analysis case
  - Right click Analysis case and then select Add.
  - Name the title as “Pseudo static” and solution type as “Slope Stability (SRM)”.
  - Click “Analysis Control” and select “strain” from Analysis Control.
  - Click OK.
  - Click Apply.



- STEP 3:
  - Right click on the Analysis case you just created as “Pseudo static” and select solve to run the analysis.
  - After the analysis go to Results > pseudo static.
  - Click on plane strain Strains
  - One can see the localization of strains by double clicking on the E-Max shear tab.
  - The Safety Factor tab shows the calculated critical factor of safety of slope.



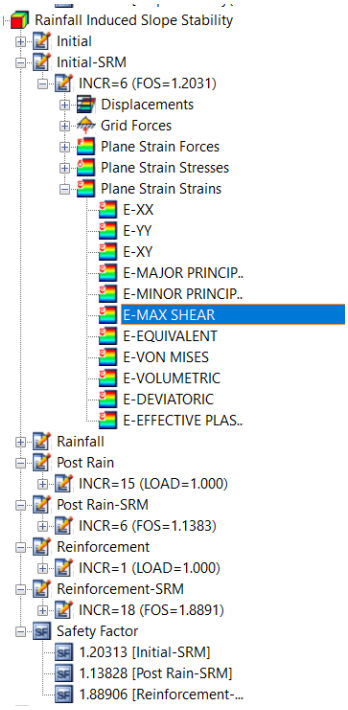
RAINFALL INDUCED SLOPE STABILITY

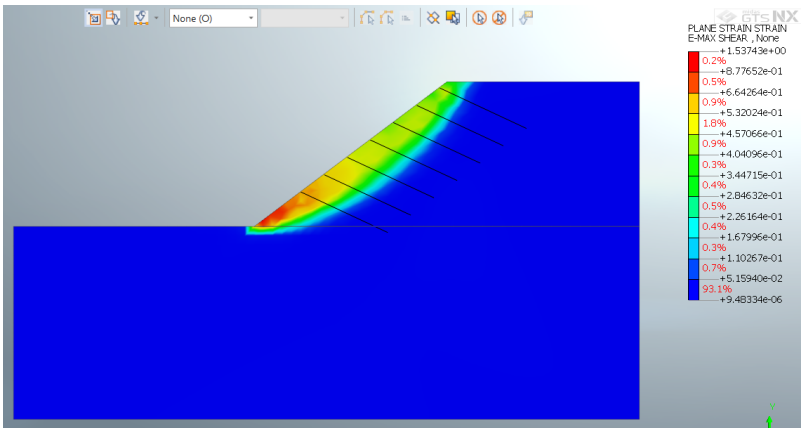
- STEP 1:
  - Go to Analysis>Perform.
  - Select “Rainfall Induced Slope Stability” from GTS NX solver.
  - Click Run.

- STEP 2:
  - Go to Results > Rainfall
  - Click on plane strain Strains>initial-SRM.

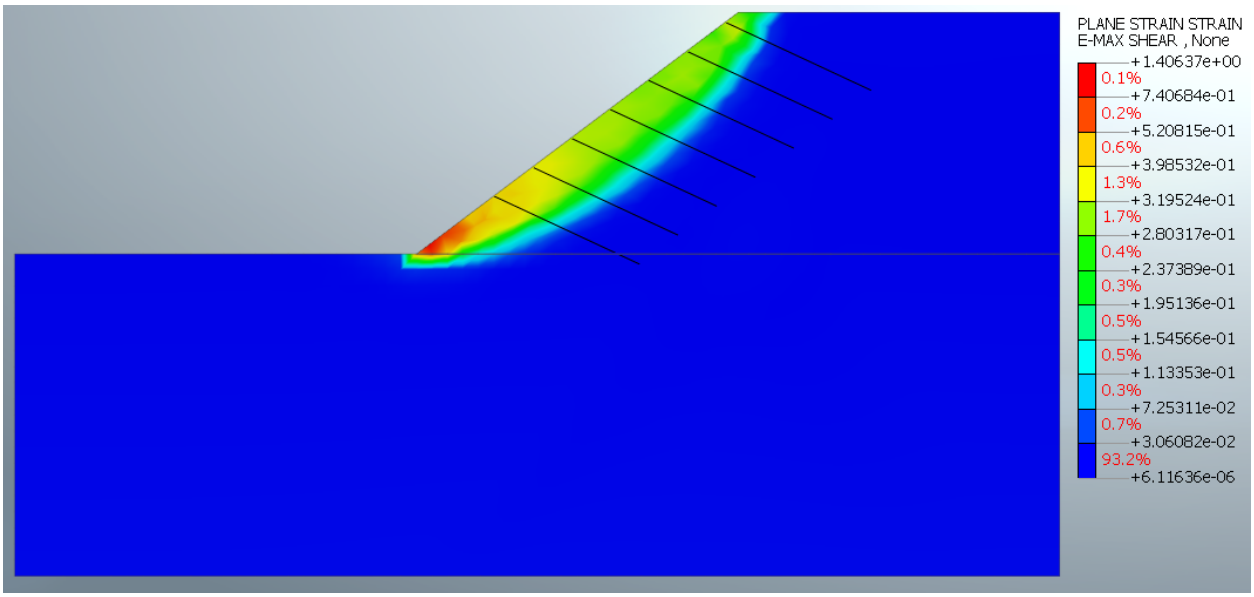
- One can see the localization of strains by double clicking on the E-Max shear tab.
- The Safety Factor tab shows the calculated critical factor of safety of slope i.e. safety factor =1.20313

Pre Rain:



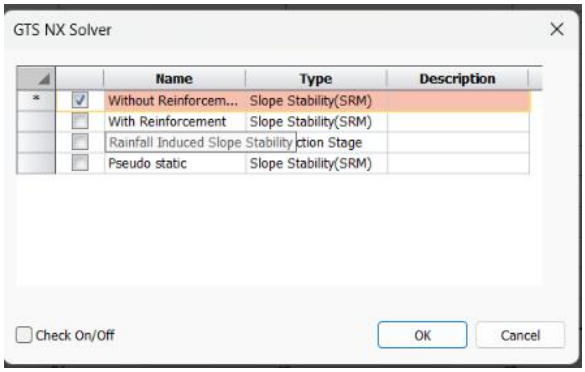


Post rain:



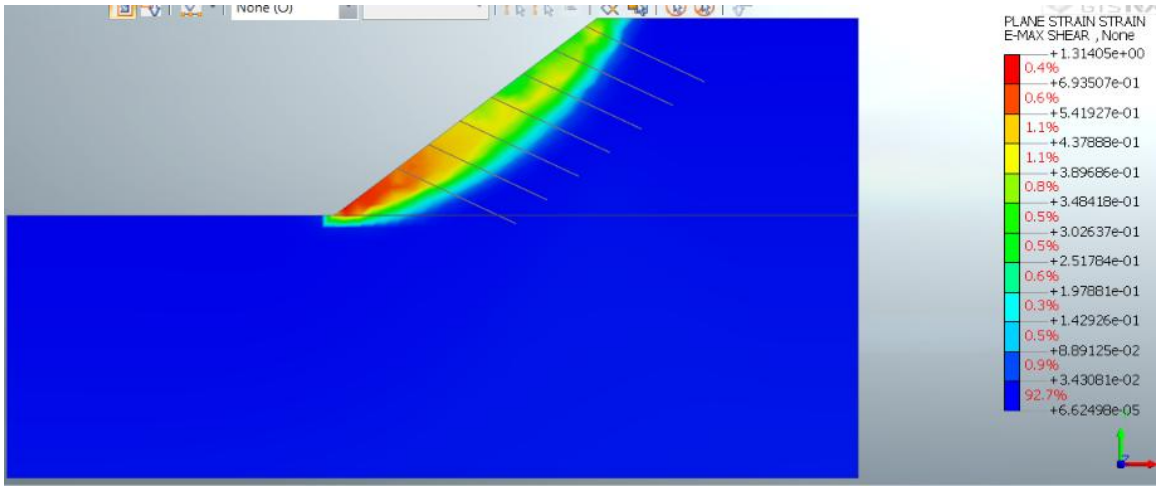
CASE 2: WITHOUT REINFORCEMENT

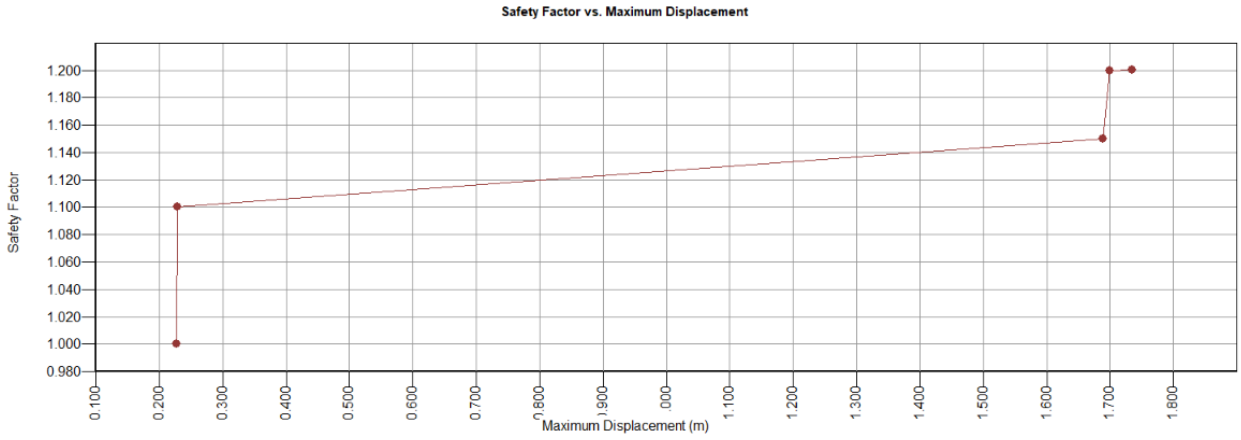
- STEP 1:
  - Go to Analysis>Perform.
  - Select “Without Reinforcement” from GTS NX solver.
  - Click Run.



- STEP 2:

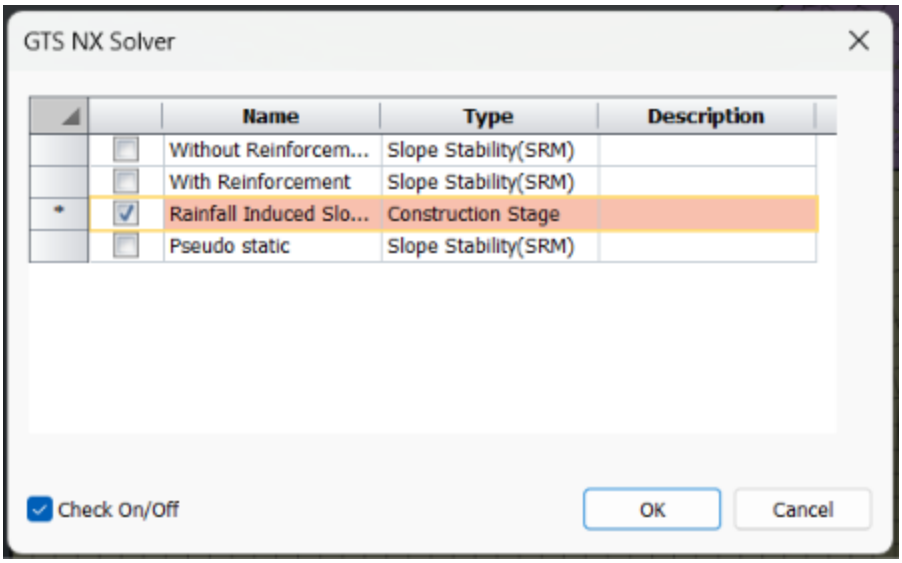
- Go to Results > Without Reinforcement
- Click on plane strain Strains
- One can see the localization of strains by double clicking on the E-Max shear tab.
- The Safety Factor tab shows the calculated critical factor of safety of slope i.e. safety factor = 1.20078



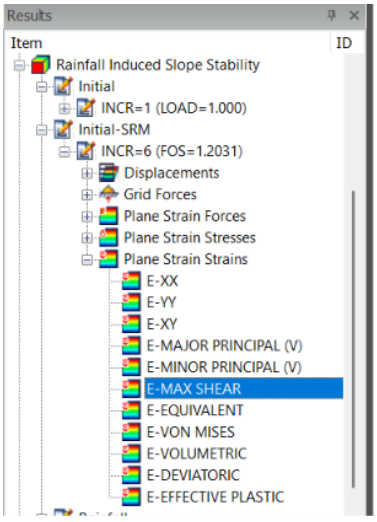


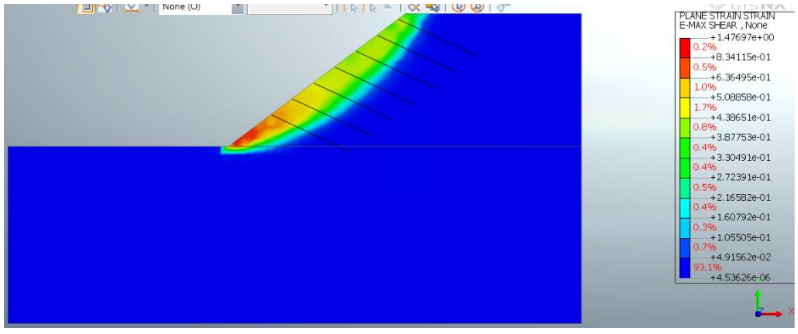
RAINFALL INDUCED SLOPE STABILITY

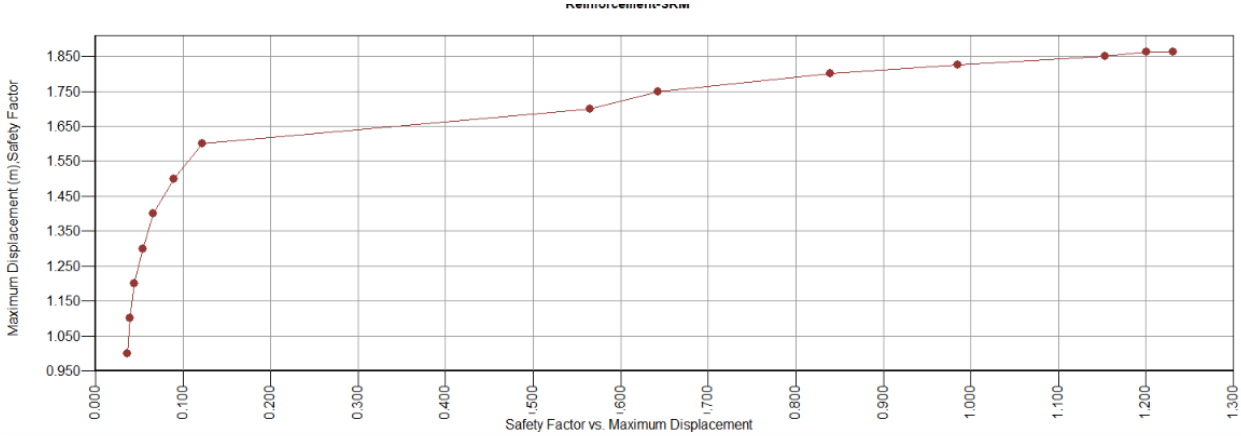
- STEP 1:
  - Go to Analysis>Perform.
  - Select “Rainfall Induced Slope Stability” from GTS NX solver.
  - Click Run.



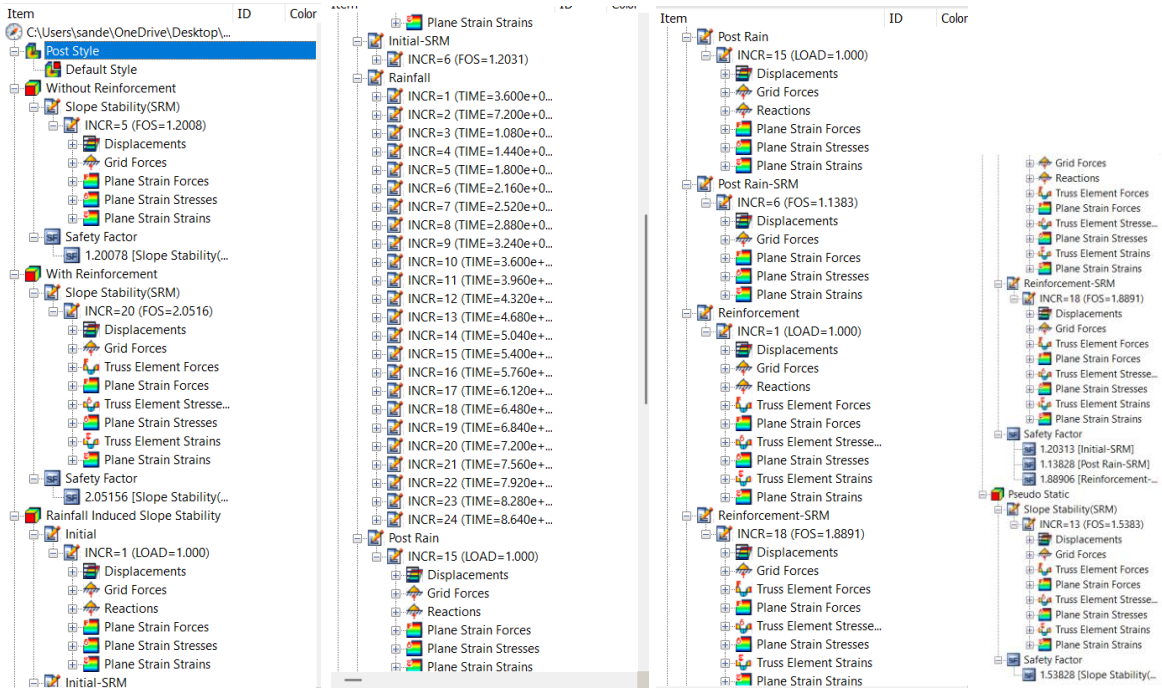
- STEP 2:
  - Go to Results > Rainfall
  - Click on plane strain Strains>initial-SRM.
  - One can see the localization of strains by double clicking on the E-Max shear tab.
  - The Safety Factor tab shows the calculated critical factor of safety of slope i.e. safety factor =1.20313







OBSERVATION:



Sl No.	Description	Safety factor
1	With Reinforcement	2.05156
2	Without Reinforcement	1.20078
3	Pseudo Static	1.53828
4	Rainfall Induced Slope Stability I. Initial-SRM II. Post rain-SRM III. Reinforcement-SRM	 I. 1.20313 II. 1.13828 III. 1.88906

CONCLUSION:

The unreinforced slope fails to meet the required operational safety threshold of 1.5, recording an initial Factor of Safety (FOS) of 1.20078. Environmental factors further compromise this stability; rainfall infiltration increases soil saturation and pore water pressure, which reduces the FOS from an initial state of 1.20313 down to a highly critical 1.13828 post-rain. Because operations cannot proceed safely under these conditions, structural intervention is strictly required.

Implementing reinforcement successfully elevates the slope's stability well above the 1.5 threshold across all tested loading conditions. By installing reinforcing elements, shear stresses are transferred from the soil to the reinforcement matrix, which intercepts potential failure planes and provides substantial resisting forces against sliding. Under standard static conditions, this reinforcement increases the FOS from 1.20078 to 2.05156.

The reinforcement also mitigates the adverse effects of environmental and dynamic loads. Following a rainfall event, the addition of reinforcement recovers the degraded stability, boosting the FOS from 1.13828 to 1.88906. Furthermore, the reinforced system demonstrates adequate resilience under pseudo-static conditions (simulating seismic loading), maintaining a safe FOS of 1.53828. Ultimately, the reinforcement actively prevents excessive slope deformation, increasing the overall shear capacity of the soil mass and ensuring the site is secure for continuous work.